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| **Fiscal Year:** | FY 2014 | **Task Last Updated:** | FY 08/30/2013 |
| **PI Name:** | Judex, Stefan Ph.D. | | |
| **Project Title:** | Genes that Predict the Loss of Bone during Weightlessness | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Musculoskeletal Biology | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
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| **Zip Code:** | 11794-5281 | **Congressional District:** | 1 |
| **Comments:** | | | |
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| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA ARC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Donahue, Leah Rae ( Jackson Laboratory ) | | |
| **Grant/Contract No.:** | NNX12AL25G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The amount of bone loss between individual astronauts is highly variable with some astronauts losing large amounts of bone while others are largely unaffected. This variability is accounted for, at least to a large extent, by genetic variations between individuals. Unfortunately, the specific genes that modulate this genetic sensitivity of bone loss in microgravity are unknown. To this end, we have recently completed a genetic study in which we identified specific regions on chromosomes (QTL or Quantitative Trait Loci) that either add to or reduce the propensity of an individual to lose bone when normal weightbearing is removed from the musculoskeleton. While providing critical information, these QTL still contain hundreds of genes, rather than a single one. Here, we propose first to greatly reduce the number of candidate genes with bioinformatics tools and then experimentally test the shortlist of candidate genes using knockout mice available from the Jackson Laboratory's (JAX) Genetic Resource Repository and the JAX Knock Out Mouse Project (KOMP) production center. The identification of single genes modulating bone loss during weightlessness can be used | | |

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| | as a critical diagnostic tool for the identification of astronauts most in need of countermeasures as well as to unravel novel drug targets that preserve bone health in space. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This proposal directly addresses one of the six highest priority recommendations made in the National Research Council's 2011 Decadal Survey Report, "Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era" ( http://www.nap.edu/ ), namely recommendation AH3: "Bone loss studies of genetically altered mice exposed to weightlessness are strongly recommended." Defining some of the specific genes that cause differences in the skeletal sensitivity to weightlessness will not only provide novel diagnostic and therapeutic means of combating bone loss in space but also on Earth in individuals subjected to bedrest or suffering from paralysis. |
| **Task Progress:** | Progress of Specific Aim 1<br>Specific Aim 1 has been largely completed. Using methods described further in the grant proposal, we performed extensive bioinformatics analyses to reduce the number of candidate genes controlling changes in trabecular BV/TV during unloading from more than 1000 to 25. Briefly, genes with clear evidence that they are not related to bone were removed. Single nucleotide polymorphisms (SNPs) that are harbored in synonymous coding regions were also removed. Genes with multiple SNPs were integrated. ConceptGen (NCIBI) with data from the Panther Pathway, Kegg Pathway, Biocarta Pathway, Gene Ontology, and MeSH was then employed to filter genes with putative functions in Wnt, Wnt-Receptor, Wnt-Protein Interactions, B-Catenin, Actin Cytoskeleton, NF-KB, TGF-B, ESR, BMP-BMPR, osteoblast, bone formation, bone resorption, and bone remodeling pathways. Inherently, this search was not exhaustive but it is likely that true regulators are among the current 25 candidate genes, particular because our gene list was also vetted against microarray data from in vitro and in vivo unloading experiments.<br><br>Progress of Specific Aim 2<br><br>Specific Aim 1 one provided a short of list of candidate genes that may modulate the magnitude of bone's response to unloading and reambulation. Our first target was CD44. In vitro studies indicate that the protein produced by this gene is associated with 'cell-cell' and 'cell-extracellular' matrix interactions (1). CD44 has also been reported in the context of osteoclast fusion and formation of multinucleated cells (2, 3). Despite the association of CD44 with osteoclast fusion, no high bone mass phenotype has been observed in CD44 KO mice (2, 4). Substrate specific CD44 roles has been suggested (2). To determine the role of CD44 in unloading induced bone loss, 26 9wk old, female B6.129 (Cg)-Cd44tm1Hbg/J (CD44-/-) and 34 C57Bl/6J (B6J) mice were assigned to three groups – baseline controls, age-matched controls and hindlimb unloaded HLU). Baseline CD44-/- (n=6) and B6J (n=10) were sacrificed at 9wk of age to determine baseline phenotypes. CD44-/- (n=10 each) and B6J (n=12 each) were subjected to 2wk of HLU (CD44_HLU and B6J_HLU) or normal cage ambulation (CD44_C and B6J_C). Ex vivo µCT scanning was performed at 10µm resolution at the distal femoral metaphysis. A 1.5 mm region proximal to the distal femoral growth plate was chosen as region of interest. Bone indices within the same genotype were tested for statistical significance using one way ANOVA and Tukey post-hoc tests. Changes in baseline bone morphology were tested for significance using a two tailed, independent samples t-test. Gene x Treatment (hindlimb unloading) interactions were analyzed using two-way ANOVA. CD44-/- mice had a baseline bone phenotype compared to wild type B6J mice with a 38% lower bone volume fraction (BV/TV), a 17% smaller trabecular number (Tb.N) and a 14% lower trabecular thickness (Tb.Th) (all $p< 0.005$). These differences persisted 2wk later when age-matched controls were compared between the two strains (all $p<0.001$). Thus, no Age x Gene interaction was observed. Two weeks of hindlimb unloading caused rapid deterioration in trabecular quantity and architecture in CD44-/- mice with 44% less BV/TV, 13% lower Tb.N, and 21% lower Tb.Th compared to age matched controls of the same genotype. Bone loss in B6J mice due to HLU was comparable to CD44-/- mice with 33% less BV/TV, 6% lower Tb.N and 19% lower Tb.Th. Because of the only small differences in the skeletal response to HLU between CD44-/- and B6J, Gene x Unloading interactions were not significant. These data highlight the importance of CD44 in acquiring trabecular morphology under normal physiological loading conditions but the contribution of CD44 towards modulating bone response to unloading is, at least based on this preliminary analysis, insignificant. We are currently testing the next two candidate genes on our shortlist.<br><br>References<br><br>1. GOODISON, S., URQUIDI, V. & TARIN, D. (1999) CD44 cell adhesion molecules, Molecular Pathology, 52, 189-196.<br><br>2. DE VRIES, T. J., SCHOENMAKER, T., BEERTSEN, W., VAN DER NEUT, R. & EVERTS, V. (2005) Effect of CD44 deficiency on in vitro and in vivo osteoclast formation, Journal of cellular biochemistry, 94, 954-966.<br><br>3. KANIA, J. R., KEHAT-STADLER, T. & KUPFER, S. R. (1997) CD44 antibodies inhibit osteoclast formation, Journal of Bone and Mineral Research, 12, 1155-1164.<br><br>4. PROTIN, U., SCHWEIGHOFFER, T., JOCHUM, W. & HILBERG, F. (1999) CD44-deficient mice develop normally with changes in subpopulations and recirculation of lymphocyte subsets, The Journal of Immunology, 163, 4917-4923. |
| **Bibliography Type:** | Description: (Last Updated: 01/16/2018) |
| **Articles in Peer-reviewed Journals** | Judex S, Zhang W, Donahue LR, Ozcivici E. "Genetic loci that control the loss and regain of trabecular bone during unloading and reambulation." Journal of Bone and Mineral Research. 2013 Jul;28(7):1537-49. http://dx.doi.org/10.1002/jbmr.1883 ; PubMed PMID: 23401066 , Jul-2013 |